

SUBTOPIC RX-ST01

OXIDATION NUMBER

OBLIGATIONS: RX-S01..S06

HARD BOUNDARY: NO TITRATION

{{ LESSON_TITLE: What is an Oxidation Number? Understand the Bookkeeping Idea Before Rules }}

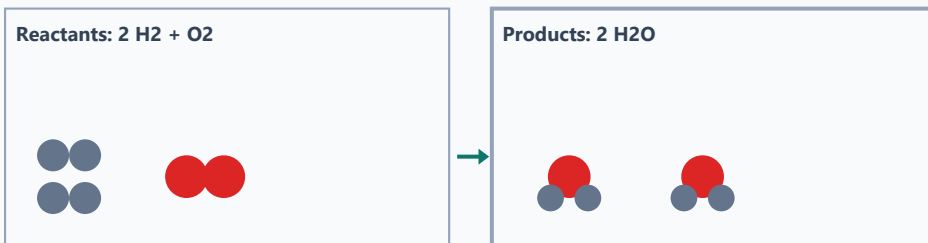
Cognitive Job: Master the bookkeeping definition before memorizing arbitrary rules. Elements have variable oxidation states.

{{ CONCEPT_FOUNDATION: The Bookkeeping Contract }}

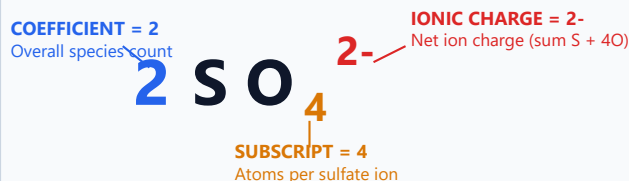
An oxidation number is NOT an actual physical charge located on an atom.

It is a hypothetical bookkeeping charge an atom would carry if all shared bonds were treated as 100% ionic, assigning electrons completely to the more electronegative atom.

- * In elemental O₂: Neither atom is more electronegative => ON = 0.
- * In H₂O: Oxygen is more electronegative than Hydrogen => H = +1, O = -2.
- * In H₂O₂ (Peroxide): The O-O bond shares electrons equally => H = +1, O = -1.
- * The Invariant: The algebraic sum of oxidation numbers must equal the net charge of the species.

{{ VISUAL_PRIMITIVE: Particulate View of Atom & Charge Conservation }}

Grey = H atom (4 total) | Red = O atom (2 total) - 100% Conserved

{{ VISUAL_PRIMITIVE: Notation Anatomy - Distinguishing Position Meaning }}**{{ CLINIC: Subscript vs Charge vs Oxidation Number }}****MISCONCEPTION CLINIC**

* **Misconception 1: 'Oxidation number is the same as valence.'**

Correction: Valence is combining capacity and is always a positive integer (e.g. Carbon valence = 4). Oxidation number is signed and can be positive, negative, zero, or fractional (e.g. CH₄: C = -4; CO₂: C = +4).

* **Misconception 2: 'In SO₄²⁻, each oxygen atom carries a 2- charge.'**

Correction: The 2- superscript is the NET CHARGE of the whole sulfate ion, not oxygen alone!

Each oxygen is assigned -2. Total oxygen charge = 4 x (-2) = -8. Sulfur is +6.

Check: +6 + (-8) = -2 = Net Ionic Charge.

* **Misconception 3: 'Oxygen is always -2.'**

Correction: In peroxides (H₂O₂, Na₂O₂), O = -1. In superoxides (KO₂), O = -1/2. In OF₂, O = +2.

WORKED EXAMPLE WITH STEP-BY-STEP AUDIT**{{ EXAMPLE: Assigning ON in K₂Cr₂O₇ and CrO₅ }}**

Problem: Determine the oxidation state of Chromium in (a) K₂Cr₂O₇ and (b) CrO₅.

Reasoning Story (Apply Fixed Rules Before Sum Rule):

Part (a) Potassium Dichromate (K₂Cr₂O₇):

- Step 1 (Fixed Rules): Group 1 metal (K) has invariant ON = +1. Potassium contribution = 2(+1) = +2.
- Step 2 (Oxygen Rule): No peroxide linkage indicated. Standard oxygen ON = -2. 7(-2) = -14.
- Step 3 (Sum Rule): Net charge = 0. Let Chromium ON = x. Formula has 2 Cr atoms:
(+2) + 2x + (-14) = 0 => 2x - 12 = 0 => 2x = +12 => x = +6.

Conclusion: In K₂Cr₂O₇, Cr = +6. [VERIFIED & AUDITED]

Part (b) Butterfly Compound (CrO₅) - The Structure-Sensitive Exception:

- 1. Naive Algebra (Wrong): If all O = -2 => x + 5(-2) = 0 => x = +10. (IMPOSSIBLE: Cr max ON = +6).**
- 2. Structural Truth:** CrO₅ contains ONE oxo oxygen (=O) and TWO peroxy linkages (-O-O-).
 - * 1 oxo oxygen with ON = -2.
 - * 4 peroxy oxygens with ON = -1 each (total -4).
- 3. Correct Sum: x + (-2) + 4(-1) = 0 => x - 6 = 0 => Cr = +6. [STRUCTURE-GROUNDED TRUTH]**

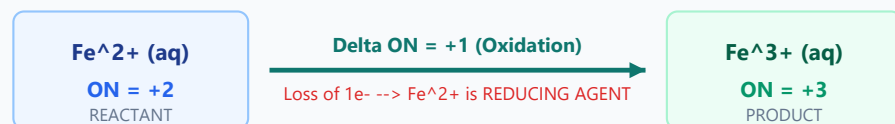
PROCESS VS AGENT

SELF / OTHER FRAME

REACTANT ROLE ATTACHMENT

{{ LESSON_TITLE: Oxidation, Reduction & the SELF vs OTHER Agent Attachment Frame }}

{{ VISUAL_PRIMITIVE: Electron Transfer & Oxidation State Progression }}



{{ RULE_PRIORITY_LADDER: The Ironclad Order of Precedence }}

Consult rules in this strict order. NEVER override an earlier rule with a later rule:

1. Priority 1 - Free Elemental State: Any uncombined element has ON = 0.

Examples: O₂, P₄, S₈, Fe, Cu, Cl₂ => all have ON = 0.

2. Priority 2 - Alkali & Alkaline Earth Metals:

Group 1 (Li, Na, K, Rb) is ALWAYS +1 in compounds. Group 2 (Be, Mg, Ca, Ba) is ALWAYS +2.

3. Priority 3 - Fluorine: The most electronegative element. In compounds, F is ALWAYS -1.

4. Priority 4 - Hydrogen: Usually +1. Exception: in metal hydrides (NaH, CaH₂), H = -1.

5. Priority 5 - Oxygen: Usually -2. Exceptions: in peroxides (H₂O₂), O = -1; in OF₂, O = +2.

6. Priority 6 - Halogens: Usually -1. Exception: when bonded to more electronegative atoms (O, F).

7. Priority 7 - The Sum Rule (Final Step):

Sum of all ONs = Net Charge of the molecule or polyatomic ion. Solve for unknown element.

SELF VS OTHER REASONING FRAME

{{ FRAME: What It Undergoes vs What It Causes }}

The central pedagogical barrier in Redox is confounding the PROCESS with the AGENT.
Remember: Agent name describes what it does to the OTHER reactant!

* **OXIDATION (Process):** Loss of electrons => Oxidation Number **INCREASES (UP)**.

* **REDUCTION (Process):** Gain of electrons => Oxidation Number **DECREASES (DOWN)**.

THE AGENT ATTACHMENT CONTRACT:

- The species that UNDERGOES OXIDATION loses electrons and donates them to the other. Therefore, it CAUSES the other species to be reduced => It is the REDUCING AGENT.
- The species that UNDERGOES REDUCTION gains electrons by taking them from the other. Therefore, it CAUSES the other species to be oxidised => It is the OXIDISING AGENT.
- Critical Scope Gate: The role belongs to the ENTIRE REACTANT SPECIES, never product!
Example: In 2Fe²⁺ + Cl₂ --> 2Fe³⁺ + 2Cl⁻, Fe²⁺ is reducing agent; Cl₂ is oxidising agent.

GUIDED DRILL & CORE 2 TRANSFER LINK

{{ GUIDED_DRILL: MnO₂ + 4HCl --> MnCl₂ + Cl₂ + 2H₂O }}

Analyze the reaction step-by-step:

* Mn in MnO₂: ON = +4 --> Mn in MnCl₂: ON = +2. (Change: -2, Reduction).

* Cl in HCl: ON = -1 --> Cl in Cl₂: ON = 0. (Change: +1, Oxidation).

* **The Oxidised Atom is: Cl⁻. The Changing Reactant Species is: HCl.**

* **Therefore, the REDUCING AGENT is: HCl. The OXIDISING AGENT is: MnO₂.**

CORE 2 TRANSFER PIPELINE LINK:

You are now qualified for the full ExamSIDE Core 2 Transfer Suite.

Target Modules: RX-C2-01 through RX-C2-24 (JEE Main 2019-2023 audited items).

Features: Primary/Supporting concept badges, H0-H3 hint ladders, verified solutions.

Next: Open Chemistry Core 2 Transfer Book.